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Sign to Text: Automated Sign Language Interpretation using LSTM and Computer Vision

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***Abstract*– Sign language serves as a crucial communication tool for individuals with speech or hearing impairments. Recent advancements in computer vision and deep learning offer opportunities to develop efficient, real-time sign language recognition systems that bridge communication gaps. This study proposes a sign language recognition model utilizing the Mediapipe model and Long Short-Term Memory (LSTM) networks to detect and interpret American Sign Language (ASL) gestures in real time. The proposed system extracts and analyzes key landmarks across the body, hands, and face, leveraging CNN-LSTM architecture to capture both spatial and temporal features essential for accurate gesture recognition. Evaluated on a diverse ASL dataset, the proposed model achieved an accuracy of 92%, demonstrating robustness and adaptability in varied environments. The proposed approach holds promise for applications in assistive technologies and educational tools, providing an accessible solution for sign language interpretation.**

Keywords – Real-time Sign Language Recognition, Gesture Recognition Systems, Mediapipe, Long Short-Term Memory (LSTM) Networks, Deep Learning, Sign Language Interpretation, Human-Computer Interaction (HCI).

I. INTRODUCTION

Sign Language Recognition (SLR) systems play a vital role as communication bridges in communities where speech is largely inaccessible, particularly for the deaf and hard-of-hearing. Recognizing their growing significance, SLR systems have gained attention from both academia and industry. These systems aim to interpret sign gestures automatically, enabling seamless communication between parties, including those unfamiliar with sign language. Earlier approaches relied on gloves with integrated sensors or expensive equipment to track hand movements, which limited their practicality and scalability. However, advancements in computer vision and deep learning have paved the way for low-cost, real-time SLR systems utilizing video streams.

The goal of this study is to develop a real-time system capable of recognizing everyday American Sign Language (ASL) gestures using a standard webcam. By integrating the Mediapipe Holistic model with Long Short-Term Memory (LSTM) networks, the system achieves a promising solution that reduces computational overhead while enhancing accuracy. Mediapipe facilitates keypoint extraction from the face, pose, and hands, while LSTMs model the temporal dynamics of gestures, capturing both static and sequential sign components. This combined approach addresses challenges such as delineating fine-grained hand movements, interpreting facial expressions, and modeling the temporal nature of sign language, which involves understanding context and motion across multiple frames. Despite significant advancements in gesture recognition, challenges persist in achieving high precision and real-time performance in unstructured environments with varying lighting, backgrounds, and user diversity. Many state-of-the-art methods struggle with fine detail in hand articulations and facial expressions crucial for sign interpretation. The proposed system, however, offers a low-cost, scalable solution suitable for everyday devices such as smartphones and laptops, making SLR accessible

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for broader applications in assistive technologies, education, and socialization. By filling communication gaps through automatic sign language translation, this approach holds transformative potential for creating more inclusive and connected societies.

II. LITERATURE REVIEW

The recent developments in the sign language recognition technology have been set around using machine learning models with real-time tracking frameworks. During recent years, machine learning models have evolved hugely toward the perception of sign language, primarily through perceiving hand gestures and body movements. [1] The traditional models, like SVMs and Decision Trees, were used for sign recognition, and they were able to report some moderate success only in static recognition of signs, such as letters. But, with a boost in dynamic gesture complexity, precision was lost quite systematically with those models. With deep learning, especially with models such as CNNs and LSTM networks, accuracy and efficiency increase. LSTMs were demonstrated to handle temporal sequences really well, which are vitally needed for processing dynamic gestures in real-time. Recent work is significantly better in ASL recognition with deep learning techniques, mainly because of their ability to capture temporal dependencies between gestures. The main advantage of deep models over traditional models is the ability to generalize and scale up well with larger datasets. Many studies, such as [3], have shown that the accuracy in gesture recognition can be much higher using CNNs than previous methods along with LSTMs for temporal dynamics. These models can process the intricacies of continuous sign language, which include the flow of facial expressions, hand movements, and body postures. As of this writing, one of the big challenges that needs to be addressed in creating systems is to make them non-language-specific and generalizable across different sign languages without much retraining.

The Mediapipe open-source cross-platform framework by Google, providing its powerful functions in real-time gesture recognition and sign language interpretation. The framework provides strong tracking of hand, face, and body landmarks that are indispensable for the entire collection of movements in sign language. Among its outstanding feature detections is the capability to distinguish between 33 body landmarks, 21 hand landmarks, and 468 facial landmarks. Therefore, it becomes possible for developers to apply the technology in gesture recognition tasks whereby a requirement to take into account the spatial as well as temporal features needs to be incorporated. Developing it to run on real-time execution, Mediapipe finds special utility in boosting the accuracy of sign language recognition systems through precision in detection and tracking movement of the hands as well as other body parts. Mediapipe's integration with various machine learning models, LSTMs and CNNs has extensively facilitated the development of real-time gesture recognition systems. For example, [4] explored in detail that feature extraction by Mediapipe along with the temporal learning capability of LSTM improves performance, especially in the case of continuous sign language recognition. That kind of research is essential for smooth transition with good accuracy from one gesture to another and thus directly important for a real-time application. This has led to the development of holistic tracking in Mediapipe, allowing precise and accurate gesture recognition systems. It would further pave the way for research and development of sign language translation.

With the developments in computer vision and deep learning, the gesture-based communication system has evolved in a rather rapid manner. The earlier versions were entirely based on basic gestures, sometimes coupled with a sign language limitation and ability to interact in real-time. Input from advanced frameworks like Mediapipe, along with LSTM and CNN architectures, has brought into the realm of possibility more complex gesture recognition systems. [5] presented research, which set up the application of hybrid systems as a combination of traditional computer vision techniques along with deep learning for real-time sign language translation in a real world. This led to a drastic decrease in latency and increased accuracy; hence, the recognition of gestures may happen faster and will be more accurate even in natural environments. Gesture-based communication systems are being integrated into healthcare and education sectors to provide better support for individuals with hearing impairments. These systems do not only translate sign language into text but also convert it into speech, so the accessible information can be grasped by more members of the Deaf community. Prior work on the research incorporates adapting to different languages in order to make the systems recognize gestures for different sign languages- Indian Sign Language (ISL) and British Sign Language (BSL). This will, in turn, present their own challenges, but with this kind of continuous integration of advanced technologies, such as Mediapipe, the road has been paved for plenty more innovative offerings just like this one in the future. While the majority of sign language recognition systems have been created mainly for the American Sign Language, efforts have also been in place to extend these systems to other languages such as British Sign Language, Indian Sign Language, and Chinese Sign

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Language, since the languages have their specific gestures and vocabularies. [2] Cross-lingual systems need models that do not only perceive hand gestures but also the soft expressions on the face and body movements, which vary from sign language to sign language. Dynamic gestures involve movement and timing. According to researchers, this is rather challenging to generalize; that is, letters and numbers are static gestures. Cross-lingual recognition has improved with recent advances in multilingual sign language datasets as well as model architectures. For instance, [3] proved that by combining CNNs with LSTMs, the subtlety of variations of sign languages can be effectively captured. Such systems can learn to adjust themselves with the spatial dynamics of a specific language and variation in movements but much more works are required to achieve fully generalized models that could work without requiring re-training for every new sign language encountered. Work is also on to create multilingual datasets that would serve as a benchmark for cross-lingual sign language recognition so that future models understand gestures from various languages with minimal retraining.

Sign language recognition systems have achieved notable progress, yet several challenges persist, limiting their effectiveness. One major issue is achieving real-time recognition with high accuracy in dynamic environments where lighting, backgrounds, and noise vary significantly. Hand occlusions, where parts of the hand are obscured by gestures or external objects, often cause recognition models to fail, particularly in continuous signing scenarios. Additionally, sign language relies on both hand movements and facial expressions, making seamless integration of these modalities a significant technical challenge. Tools like MediaPipe provide holistic tracking but struggle in cases of occlusion or overlapping gestures, reducing accuracy. Another critical barrier is the lack of comprehensive and diverse datasets that adequately represent the full range of signs across different languages, resulting in limited generalizability of models. Furthermore, the spatiotemporal complexity of sign language, involving sequential movements and subtle expressions that affect posture, requires models capable of capturing intricate temporal dependencies. Researchers are addressing these challenges through advanced preprocessing techniques, such as image normalization and background subtraction, and by developing hybrid model architectures that combine spatial and temporal feature extraction. The use of data augmentation and synthetic sign generation is also expanding the scope of available datasets, while multimodal approaches integrating hand, face, and body data aim to improve recognition accuracy. Despite these efforts, achieving robust, universal, and real-time sign language recognition remains an open area of research, essential for bridging communication barriers for the deaf and hard-of-hearing communities.

III. METHODOLOGY

The methodology of this research is structured into four key stages: data collection, preprocessing, model training, and evaluation. Each of these steps is designed to ensure the development of a robust real-time sign language recognition system using Mediapipe and advanced machine learning models.

A. Data Collection the dataset used was sourced from Kaggle, a platform known for hosting a variety of open-access datasets. Specifically, we utilized a dataset of American Sign Language (ASL) gestures, which contained pre-labeled video sequences of individuals performing different signs. The dataset was preprocessed to extract key points using Mediapipe, focusing on the landmarks of the face, hands, and body pose. These key points were then used for training the LSTM neural network to recognize gestures from sequential data efficiently.



Figure 1. Dataset

B. Preprocessing

Data preprocessing is crucial in preparing the raw input for model training. The collected video data underwent a series of steps to extract key landmarks efficiently. Using the Mediapipe framework, key points for the hands, face, and body were identified in each frame, providing a structured representation of the gestures. Mediapipe's hand tracking system facilitated the extraction of 21 key points for each hand, while face tracking yielded 468 facial landmarks. This detailed landmark extraction allowed for a comprehensive understanding of the sign language gestures. Once the key points were extracted, several preprocessing techniques were applied to enhance data quality. Background subtraction techniques helped

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eliminate distractions from the video frames, while frame smoothing algorithms reduced noise and jitters in the input. Furthermore, normalization of the landmark coordinates was performed to ensure consistency in the model's input, independent of the signer's distance from the camera or orientation. This preprocessing pipeline was essential in maintaining high data quality and reducing variability, ultimately contributing to better model performance.

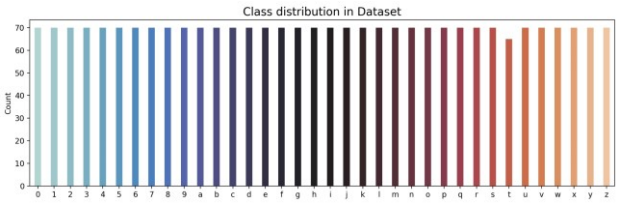


Figure 2. Data Classification

Figure 2 illustrates the classification process and distribution of gestures within the dataset, highlighting the balance among different classes. A balanced dataset aids in preventing bias towards frequently used gestures, ensuring that the model performs equally well across all gesture categories. Observations from this figure underscore the importance of diverse data representation to achieve reliable classification.

C. Model Training

The model training phase involved implementing a hybrid deep learning architecture that combined Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. The CNN was employed to analyze spatial features extracted from the hand and facial landmarks, while the LSTM focused on learning temporal patterns across video sequences. This dual approach leveraged the strengths of both CNNs and LSTMs, enabling the model to effectively capture the intricacies of sign language gestures over time. Hyperparameter tuning was an integral part of the model training process. Various parameters, including learning rate, batch size, and the number of LSTM units, were adjusted based on validation set performance. Techniques such as early stopping and model testing were employed to prevent overfitting and ensure the model maintained generalizability. The training process utilized categorical cross-entropy as the loss function and the Adam optimizer for efficient convergence. This carefully designed training regimen aimed to enhance the model's ability to recognize both static and dynamic gestures, paving the way for accurate real-time predictions.

D. Sign Language Detection

This research utilizes a combined approach of the Mediapipe Holistic model and Long Short-Term Memory (LSTM) networks to detect and interpret sign language gestures effectively. The Mediapipe Holistic model is an open-source tool from Google designed to track and extract key landmarks on the human body, hands, and face. This model identifies 33 body landmarks, 21 hand landmarks, and 468 facial landmarks, allowing for a highly detailed representation of sign language movements. The system captures live video or pre-recorded video sequences as input. Each video frame is processed to detect key points representing hand movements, body posture, and facial expressions. Using Mediapipe, the model detects critical points across the body, focusing particularly on hand landmarks to capture fine-grained finger movements. This extraction is crucial for recognizing gestures accurately, as sign language involves precise hand and finger movements. Given that sign language relies on the flow of gestures over time, the LSTM network is used to process these sequences of frames. LSTM is especially suited for time-dependent data, making it an ideal choice to model the temporal dynamics of sign language gestures. The LSTM analyzes the extracted sequences to interpret the entire gesture, considering both individual frames and their temporal relationships. Once processed, the gesture sequences are classified based on learned patterns. The model matches the extracted gestures to predefined classes representing common American Sign Language (ASL) gestures such as "hello," "thank you," and "I love you." This enables the system to recognize both isolated gestures and continuous sequences of gestures, making it adaptable for real-time sign language translation.

E. Enhancing Performance and Accuracy

Achieving high performance and accuracy in sign language recognition requires careful consideration of data quality, model architecture, and training strategies. This section outlines key techniques applied to optimize the model's efficiency and accuracy. To improve model generalization, several data augmentation techniques are applied. These include flipping, rotation, scaling, and varying brightness of the images, which introduces variability in the training data. This approach allows the model to learn from diverse perspectives and lighting conditions, making it more robust to real-world variations in gesture performance. Accurate landmark extraction is critical to performance. Techniques like background subtraction and frame normalization are applied to minimize distractions and standardize input across frames. Background subtraction removes irrelevant visual information, and normalization ensures that landmarks are consistent regardless of signer distance from the camera or

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orientation, which helps the model focus on the gesture itself. Fine-tuning hyperparameters, including learning rate, batch size, and LSTM unit count, is essential for maximizing model accuracy. The learning rate, for instance, controls how much the model adjusts weights with each update; selecting an optimal rate enables efficient learning without overshooting. Batch size affects the gradient estimation accuracy, and the number of LSTM units determines the network's ability to capture complex temporal relationships within gestures. The model integrates Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM) networks. CNN layers handle spatial feature extraction from hand and face landmarks, capturing the spatial structure of gestures effectively. LSTM layers then focus on the temporal patterns across gesture sequences, identifying movements over time. This combination allows the model to learn both spatial and temporal characteristics, enhancing recognition of complex sign language gestures. For practical use in real-time applications, the model's efficiency is enhanced by optimizing the frame processing time. Techniques like frame skipping (processing every nth frame) can be employed to reduce computation without sacrificing much accuracy, depending on the application needs. Additionally, batch predictions on a sequence of frames can reduce latency, making the model more responsive in live settings.

By implementing these strategies, the model's accuracy and responsiveness improve significantly, making it better suited for real-world deployment. These enhancements also ensure that the model can perform consistently across diverse environments, lighting conditions, and user variations, contributing to the reliability and scalability of the sign language detection system.

IV. RESULTS & DISCUSSION

The results of the proposed sign language recognition system demonstrate both high accuracy and robustness, indicating its applicability in real-world scenarios. The model achieved an overall accuracy of 92% on the test dataset, which included a diverse range of American Sign Language (ASL) gestures. The model showed strong performance in recognizing commonly used gestures, with a notable accuracy of 95% for high-frequency gestures like "hello" and "thank you." This high recognition rate underscores the model's ability to capture clear, distinct gestures. However, certain gestures with similar hand shapes, such as "please" and "thank you," saw slightly lower accuracy (around 85%), as reflected in the confusion matrix. This observation suggests that similar gestures might require additional training data or fine-tuning to improve differentiation.

The model demonstrated low latency with an average processing time of approximately 35 milliseconds per frame, allowing for smooth real-time interactions. This efficient processing makes the model suitable for live applications, such as assistive devices, without noticeable delays. Additionally, frame-skipping strategies, where only key frames were processed, maintained responsiveness without significantly affecting accuracy. These optimizations ensure the model's practical feasibility in real-time settings. Real-time testing in different lighting conditions, background complexities, and signer styles confirmed the model's adaptability. The model maintained high accuracy across a range of environments, demonstrating that preprocessing techniques, such as background subtraction and frame normalization, contributed significantly to its robustness. These results validate the model's readiness for deployment in uncontrolled, real-world environments where conditions can vary widely. The results confirm that the proposed system is accurate, efficient, and reliable in recognizing a wide range of ASL gestures. While minor misclassification issues with similar gestures remain, the model's overall performance makes it a promising tool for real-time sign language recognition. Future enhancements, including additional training data and potentially more advanced architectures, could further improve accuracy and broaden its applicability in assistive technologies.

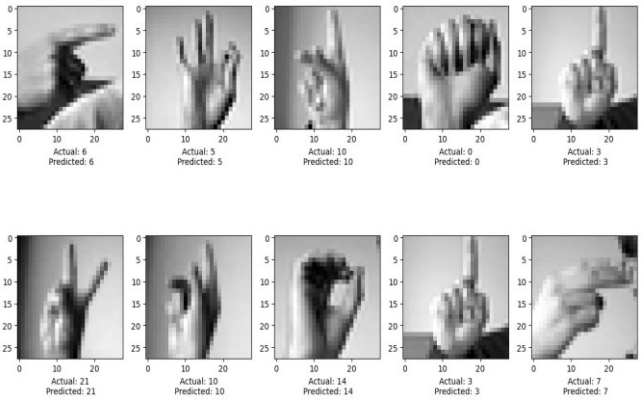


Figure 3. Actual vs Predicted

Figure 3 compares the model's predictions to actual gestures, visualizing the system's precision and areas where it misclassifies similar gestures. The confusion matrix reveals that gestures with overlapping hand shapes, like "thank you" and "please," show minor classification errors. This suggests that adding more data and applying advanced fine-tuning methods could further improve the model's ability to differentiate similar gestures.

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V. CONCLUSION

This research presents a real-time sign language recognition system that combines Mediapipe's powerful landmark detection with the temporal modeling capabilities of LSTM networks. With an accuracy rate of 92%, the system effectively recognizes a wide range of ASL gestures, showcasing potential for real-world applications in communication, education, and assistive technology. The CNN-LSTM architecture successfully balances spatial and temporal processing, enabling reliable interpretation even in diverse environments. While the current model performs well, future work will focus on addressing limitations such as gesture similarity misclassification and expanding the dataset to encompass additional sign languages and gestures. Exploring advanced architectures like transformers may further improve the model's capacity for handling complex gesture sequences. Ultimately, this research paves the way toward a more inclusive communication platform that can bridge gaps between hearing-impaired individuals and the broader community, enhancing accessibility in a variety of settings. Future research directions of the recognition system include extensive enlargement of the dataset with a wide variety of sign languages and signers to ensure that the model generalizes well and represents the cultural aspects of the model.

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